

7 Days vs. 14 Days of Antibiotics for Bloodstream Infections

Critical Appraisal of the BALANCE Trial

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RCT

THE NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Antibiotic Treatment for 7 versus 14 Days in Patients with Bloodstream Infections

The BALANCE Investigators, for the Canadian Critical Care Trials Group, the Association of Medical Microbiology and Infectious Disease Canada Clinical Research Network, the Australian and New Zealand Intensive Care Society Clinical Trials Group, and the Australasian Society for Infectious Diseases Clinical Research Network

ABSTRACT

BACKGROUND

Bloodstream infections are associated with substantial morbidity and mortality. Early, appropriate antibiotic therapy is important, but the duration of treatment is uncertain.

METHODS

In a multicenter, noninferiority trial, we randomly assigned hospitalized patients (including patients in the intensive care unit [ICU]) who had bloodstream infec-

The authors' full names, academic degrees, and affiliations are listed at the end of the article. Dr. Daneman can be contacted at nick.daneman@sunnybrook.ca or at the Division of Infectious Diseases, H-wing, Rm. 259, Sunnybrook Health Sciences Centre, University of Toronto, 2075 Bayview Ave., Toronto, Ontario M4N

Trial Identity and Citation

TRIAL NAME & CITATION

BALANCE Trial: Bacteremia Antibiotic Length Actually Needed for Clinical Effectiveness

Citation: The BALANCE Investigators. N Engl J Med 2025;392:1065-78. doi:10.1056/NEJMoa2404991

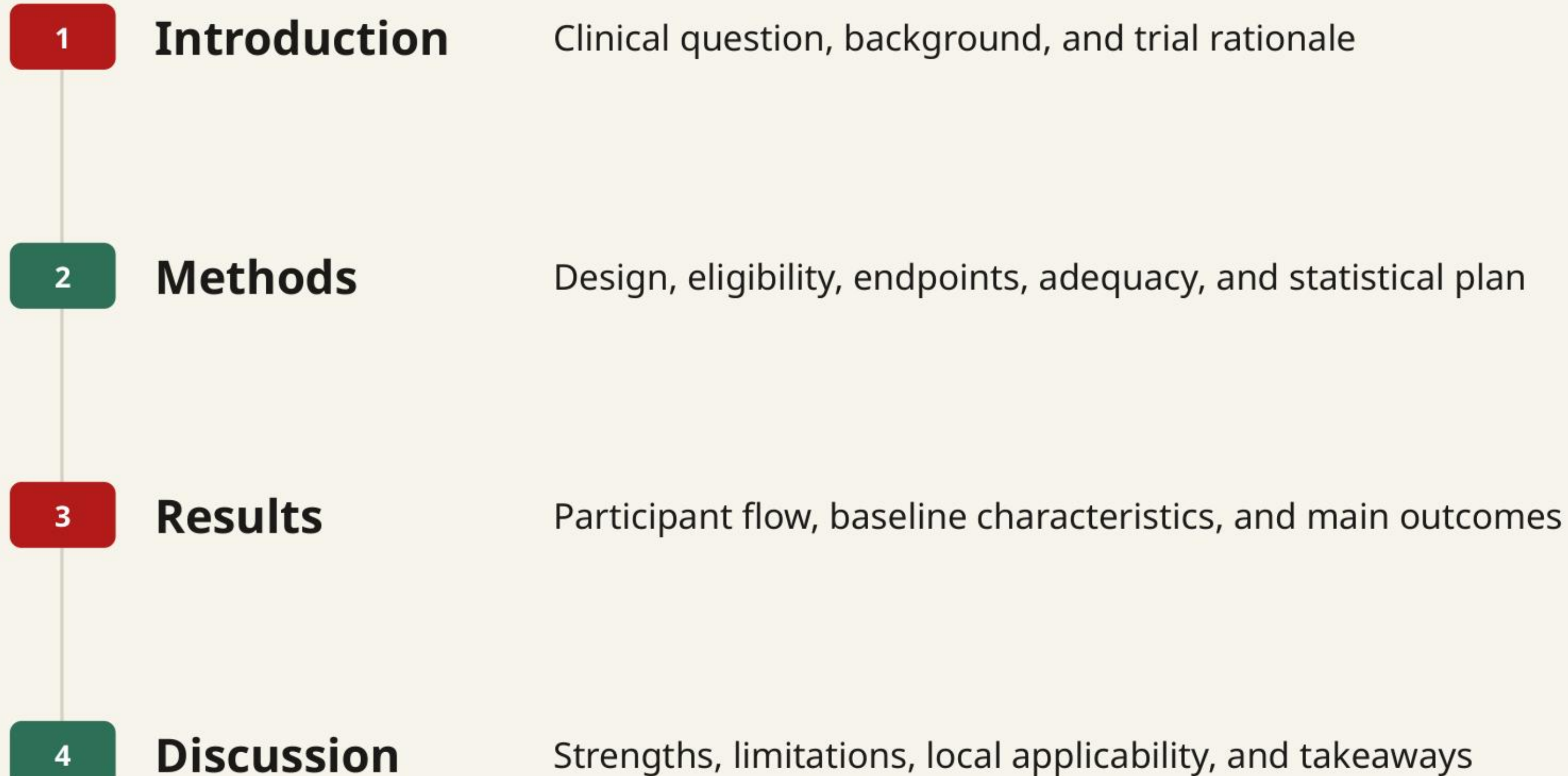
FUNDING & RESEARCH GROUPS

- Funded by the Canadian Institutes of Health Research (CIHR).
- Co-funded by national health research councils in Australia and New Zealand.
- Conducted by established academic critical care and trials groups.

TRIAL REGISTRATION & SPONSORS

- Prospectively registered at ClinicalTrials.gov under number NCT03005145.
- Coordinated by the Canadian Critical Care Trials Group and ANZICS Clinical Trials Group.
- Academic sponsorship with absolute independence and zero pharmaceutical involvement.

Structure of This Appraisal



INTRODUCTION

Background, Clinical Need, and Trial Rationale

Why This Paper Matters

Clinical problem: Traditional Dogma: Bloodstream infections have historically been treated for 14 days or longer out of fear of relapse.

Evidence gap: Lack of Evidence: Treatment durations were largely based on observational data and consensus opinions.

Research question: Stewardship Need: Unnecessary exposure to broad-spectrum antibiotics drives systemic resistance and toxicity.

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Clinical Question & PICO Framework

- P** — Hospitalized adults (ward or ICU) with a pathogenic bacterium in at least one blood culture bottle.
- I** — 7 days of adequate (susceptible-active) antibiotic therapy.
- C** — 14 days of adequate (susceptible-active) antibiotic therapy.
- O** — All-cause mortality by 90 days after index culture collection.

METHODS

Study Population, Eligibility, Outcomes, and Statistics

Study Design & Methodology

TRIAL ARCHITECTURE

- Trial Type: Multicenter, open-label, randomized, controlled, noninferiority trial.
- Randomization: Web-based 1:1 assignment, stratified by hospital site and ICU vs. ward status, using variable block sizes.

BIAS CONTROLS

- Blinding: Open-label after Day 7, but outcome assessors and resistance adjudicators were fully blinded.
- Allocation Concealment: Concealed from clinicians until Day 7 of adequate therapy, minimizing early co-intervention bias.

Eligibility Criteria

KEY INCLUSION CRITERIA

- Key Inclusions: Hospitalized patients with positive blood cultures for a pathogen.

KEY EXCLUSION CRITERIA

- Key Pathogen Exclusions: Staphylococcus aureus, Staphylococcus lugdunensis, Candida/fungi, or rare organisms.
- Key Syndrome Exclusions: Endocarditis, osteomyelitis, septic arthritis, undrained abscesses, or unremoved prosthetics.
- Patient Exclusions: Severe immunosuppression (neutropenia < 500/uL, post-transplant immunosuppressants).
- Contaminants: Single positive culture with common skin contaminants (e.g., CoNS).

Crucial Exclusion: 7-day course is NOT validated for Staphylococcus aureus, fungemia, endocarditis, or severe neutropenia.

Antibiotic Adequacy & Stewardship Rules

- Adequacy Definition: Antibiotic therapy to which the culprit pathogen was susceptible according to local lab reports.
- Clinical Discretion: Choice of antibiotic agent, dose, frequency, and IV-to-oral switch left entirely to treating team.
- Pragmatic Design: Reflects real-world clinical choices and allows variable practice within both arms.
- Stewardship Integration: Encourages switching from IV to oral route once clinical stability is achieved.

Trial Endpoints & Definitions

PRIMARY ENDPOINT

- All-cause mortality by 90 days after index culture collection.
- Objective, unambiguous, patient-important endpoint.
- Chosen to capture late relapses or therapy complications.

SECONDARY CLINICAL

- ICU mortality and hospital mortality.
- Relapse of bacteremia (positive blood culture with same organism).
- Adjudicated by a blinded registry panel.

SAFETY & RESOURCES

- Clostridioides difficile infection rates.
- Severe drug allergies, anaphylaxis, AKI, or acute drug hepatitis.
- Hospital & ICU length of stay.
- Ventilator-free, vasopressor-free, and antibiotic-free days.

Statistical Plan & Margin Choice

4.0%

noninferiority margin

3,626

target enrollment

80%

statistical power

95.7%

confidence interval

- Noninferiority Margin: 4.0 percentage points, a conservative threshold compared to standard 10% margins.
- Power Calculations: Sample size of 3,626 patients provided 80% power at one-sided alpha of 2.5%, assuming 22% mortality.
- Alpha spending: 95.7% confidence intervals used to adjust for alpha spending during three interim looks.
- Analysis Populations: Primary ITT population, with supportive per-protocol and modified ITT sensitivity checks.

Methods Critical Appraisal: Internal Validity

METHODOLOGICAL STRENGTHS

- Methodological Triumph: Allocation concealment until Day 7 prevented early differences in supportive care.
- Outcome Assessor Blinding: Primary outcome (90-day mortality) and resistant pathogen adjudication blinded, protecting against bias.
- Pragmatic Enrollment: Variable block sizes and multicenter design minimize selection bias.

LIMITATIONS & CONCERNS

- Blinding Limitation: Clinicians were not blinded after Day 7, which could theoretically influence discharge timing.

Overall CASP Verdict: High-quality, pragmatic trial with robust randomization and blinded outcome adjudication.

RESULTS

Participant Flow, Baseline comparability, and Outcomes

Figure 1: Participant Flow Diagram

Screening, Enrollment, and Follow-up (BALANCE CONSORT)

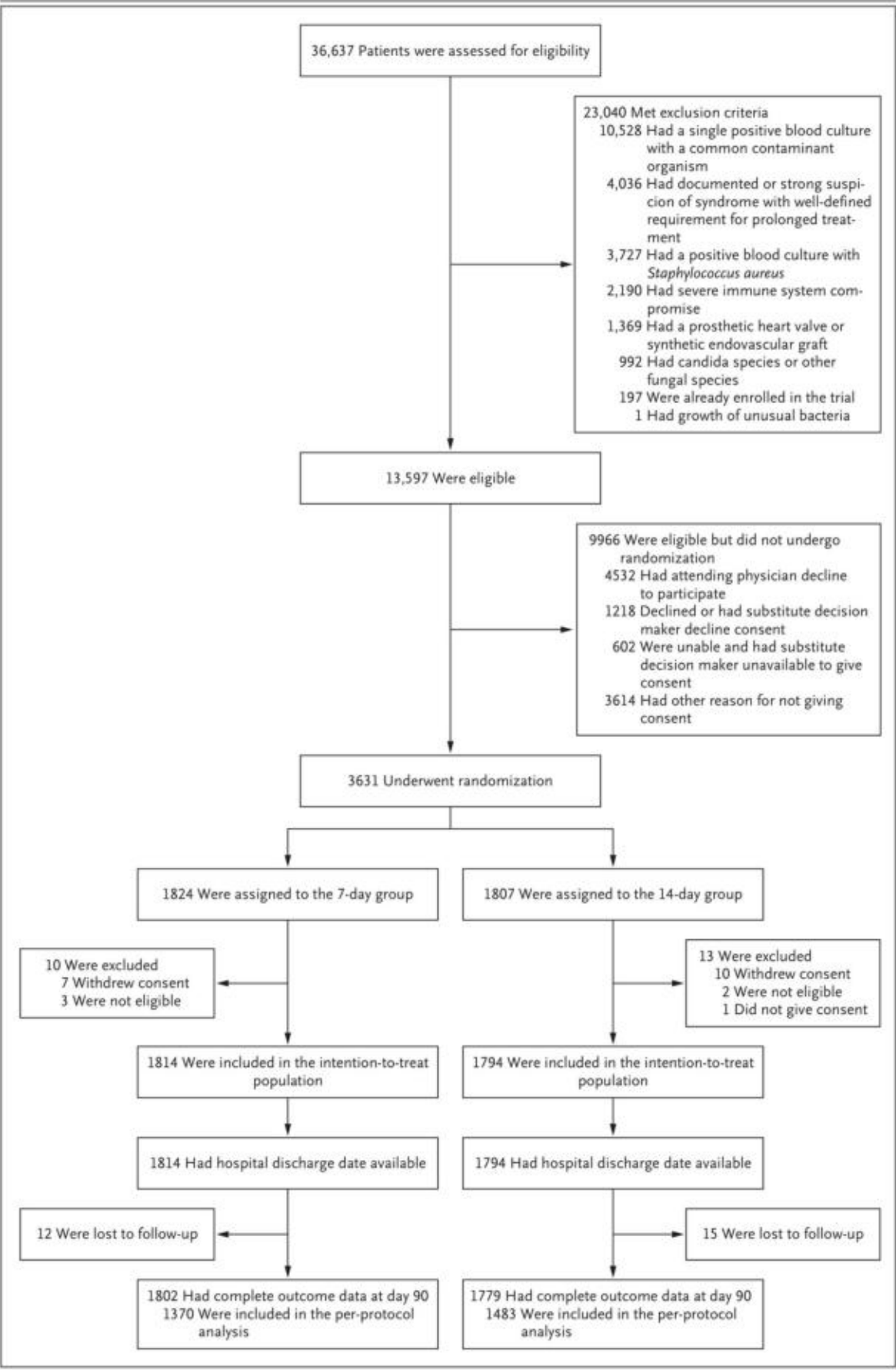


Figure 1. Screened, eligible, randomized, and analyzed patient flow. Note the high retention (99.3%).

Table 1: Baseline Patient Characteristics

Demographics and Illness Severity (BALANCE)

Table 1. Characteristics of the Patients, Infections, and Pathogens at Baseline (Primary Intention-to-Treat Analysis).*			
Characteristic	Overall (N = 3608)	7-Day Group (N = 1814)	14-Day Group (N = 1794)
Male sex — no. (%)	1922 (53.3)	974 (53.7)	948 (52.8)
Median age (IQR) — yr	70 (59–80)	70 (58–80)	70 (59–80)
Median SOFA score on day 0 (IQR)†	4 (2–8)	4 (2–8)	5 (2–8)
Enrolled in ICU — no. (%)	1986 (55.0)	997 (55.0)	989 (55.1)
Enrolled in hospital ward — no. (%)	1622 (45.0)	817 (45.0)	805 (44.9)
Receiving mechanical ventilation — no. (%)	766 (21.2)	374 (20.6)	392 (21.9)
Coexisting conditions — no. (%)			
Diabetes mellitus	1148 (31.8)	596 (32.9)	552 (30.8)
Solid-organ cancer	782 (21.7)	400 (22.1)	382 (21.3)
Obesity	655 (18.2)	331 (18.2)	324 (18.1)
Arrhythmia	540 (15.0)	264 (14.6)	276 (15.4)
Glucocorticoid use or immunosuppression‡	440 (12.2)	230 (12.7)	210 (11.7)
Chronic obstructive pulmonary disease	393 (10.9)	198 (10.9)	195 (10.9)
Renal insufficiency	425 (11.8)	217 (12.0)	208 (11.6)
Coronary artery disease	393 (10.9)	193 (10.6)	200 (11.1)
Congestive heart failure	386 (10.7)	205 (11.3)	181 (10.1)
Liver disease	227 (6.3)	117 (6.4)	110 (6.1)
Peripheral vascular disease	223 (6.2)	107 (5.9)	116 (6.5)
Dialysis dependency	127 (3.5)	60 (3.3)	67 (3.7)
Leukemia or lymphoma	101 (2.8)	49 (2.7)	52 (2.9)
Median Clinical Frailty Scale score (IQR)§	4 (3–5)	4 (3–5)	4 (3–5)
Any use of procedures to control the source of infection — no. (%)¶	1621 (44.9)	795 (43.8)	826 (46.1)
Source of acquisition of bacteremia — no. (%)			
Community	2722 (75.4)	1380 (76.1)	1342 (74.8)
Hospital ward	483 (13.4)	231 (12.7)	252 (14.0)
ICU	403 (11.2)	203 (11.2)	200 (11.1)
Source of bacteremia — no. (%)			
Urinary tract	1523 (42.2)	757 (41.7)	766 (42.7)
Intraabdominal or hepatobiliary	679 (18.8)	337 (18.6)	342 (19.1)
Lung	469 (13.0)	229 (12.6)	240 (13.4)
Vascular catheter	229 (6.3)	116 (6.4)	113 (6.3)
Skin, soft tissue, or both	187 (5.2)	104 (5.7)	83 (4.6)
Other	67 (1.9)	37 (2.0)	30 (1.7)
Undefined or unknown	454 (12.6)	234 (12.9)	220 (12.3)
Most commonly isolated pathogens in blood cultures — no. (%)			
<i>Escherichia coli</i>	1582 (43.8)	805 (44.4)	777 (43.3)
<i>Klebsiella</i> species	552 (15.3)	273 (15.0)	279 (15.6)
<i>Enterococcus</i> species	250 (6.9)	119 (6.6)	131 (7.3)
Coagulase-negative staphylococci	174 (4.8)	81 (4.5)	93 (5.2)
<i>Pseudomonas</i> species	170 (4.7)	80 (4.4)	90 (5.0)

Table 1. Demographics, ICU status, comorbidities, acquisition source, and infection foci at baseline.

Table 1 (Continued): Pathogen Profile

Pathogens Isolated at Baseline (BALANCE)

Table 1. (Continued.)			
Characteristic	Overall (N = 3608)	7-Day Group (N = 1814)	14-Day Group (N = 1794)
<i>Streptococcus pneumoniae</i>	164 (4.5)	86 (4.7)	78 (4.3)
Enterobacter species	157 (4.4)	80 (4.4)	77 (4.3)
Proteus species	133 (3.7)	58 (3.2)	75 (4.2)
Serratia species	86 (2.4)	38 (2.1)	48 (2.7)
<i>S. pyogenes</i>	74 (2.1)	39 (2.1)	35 (2.0)
<i>S. agalactiae</i>	75 (2.1)	40 (2.2)	35 (2.0)
Number and type of organisms — no. (%)			
Monomicrobial, gram-negative	2562 (71.0)	1299 (71.6)	1263 (70.4)
Monomicrobial, gram-positive	625 (17.3)	323 (17.8)	302 (16.8)
Polymicrobial	421 (11.7)	192 (10.6)	229 (12.8)

* ICU denotes intensive care unit and IQR interquartile range.
† Scores on the Sequential Organ Failure Assessment (SOFA) range from 0 to 24, with higher scores indicating more severe organ failure.
‡ Immunosuppression included chemotherapy and prednisone or equivalent glucocorticoid use of more than 15 mg per day.
§ Scores on the Clinical Frailty Scale range from 1 to 9, with higher scores indicating greater frailty.
¶ One patient in the 14-day group was missing source-control data.
|| See the Supplementary Appendix for the full list of organisms.

Baseline Comparability Critique

- Demographic Balance: Median age (70 yrs), male sex (53.3%), and Clinical Frailty Scale (4) well balanced.
- Severity Balance: Enrolled in ICU (55.0%), mechanical ventilation (21.2%), and median SOFA score (4) balanced.
- Infection Foci: Urinary tract (42.2%), intraabdominal (18.8%), lung (13.0%), and catheter (6.3%) well balanced.
- Microbiology Balance: Monomicrobial Gram-negatives (71.0%), Gram-positives (17.3%), and polymicrobial (11.7%) balanced.
- Source Control: Procedures to control the source of infection performed in 44.9% (balanced between arms).

Verdict: Randomisation succeeded. Baseline characteristics, ICU status, organ support, and pathogens were exceptionally well balanced between the 7-day and 14-day arms.

Table 2: Trial Results - Primary & Secondary Outcomes

90-Day Mortality and Clinical Measures (BALANCE)

Table 2. Primary and Secondary Outcomes.			
Outcome	7-Day Group (N=1814)	14-Day Group (N=1794)	Difference (95% CI)* percentage points
Primary outcome, death from any cause by 90 days — no./total no. (%)			
Primary analysis, intention-to-treat population	261/1802 (14.5)	286/1779 (16.1)	−1.6 (−4.0 to 0.8)
Secondary analysis, per-protocol population	178/1370 (13.0)	222/1483 (15.0)	−2.0 (−4.5 to 0.6)
Modified intention-to-treat analysis, survival ≥7 days	247/1788 (13.8)	272/1765 (15.4)	−1.6 (−3.9 to 0.7)
Secondary outcomes			
Death in hospital — no. (%)†	168 (9.3)	184 (10.3)	−1.0 (−2.9 to 0.9)
Death in ICU — no./total no. (%)‡	91/1014 (9.0)	97/1008 (9.6)	−0.6 (−3.2 to 1.9)
Median no. of days in hospital (IQR)	10 (6–21)	11 (6–22)	−1 (−1.5 to −0.5)
Median no. of hospital-free days by day 28 (IQR)	17 (0–21)	15 (0–21)	2 (0.8 to 3.2)
Median no. of days in ICU (IQR)§	5 (3–11)	5 (3–11)	0 (−0.4 to 0.4)
Median no. of days of vasopressor use (IQR)¶	3 (2–5)	3 (2–4)	0
Median no. of days of mechanical ventilation (IQR)	6 (3–14)	5 (2–12)	1 (−0.6 to 2.6)
Relapse of bacteremia — no. (%)	47 (2.6)	39 (2.2)	0.4 (−0.6 to 1.4)
Median no. of antibiotic-free days by day 28 (IQR)**	19 (11–21)	14 (11–14)	5 (4.6 to 5.4)
Antimicrobial-related adverse outcomes — no. (%)			
Allergy	14 (0.8)	19 (1.1)	−0.3 (−0.9 to 0.3)
Anaphylaxis	1 (0.1)	1 (0.1)	0 (−0.2 to 0.2)
Acute kidney injury	15 (0.8)	17 (0.9)	−0.1 (−0.7 to 0.5)
Acute hepatitis	2 (0.1)	4 (0.2)	−0.1 (−0.4 to 0.2)
<i>Clostridioides difficile</i> infection — no. (%)	31 (1.7)	35 (2.0)	−0.2 (−1.1 to 0.6)
Secondary infection or colonization with antibiotic-resistant organisms — no. (%)	173 (9.5)	152 (8.5)	1.1 (−0.8 to 2.9)
Secondary infection or colonization with antibiotic-resistant organisms in sterile culture — no. (%)	20 (1.1)	24 (1.3)	−0.2 (−1 to 0.5)
* Differences are expressed as absolute risk differences or, for variables shown as medians, as median differences. A 95.7% confidence interval is shown for the primary analysis (accounting for alpha spending in interim analyses), and 95% confidence intervals are shown for the per-protocol analysis, the modified intention-to-treat analysis, and the secondary outcomes. The widths of the confidence intervals for secondary outcomes have not been adjusted for multiplicity. The 95% confidence intervals for the median differences were estimated with the use of quantile regression.			
† One patient in the 7-day group is still in the hospital.			
‡ Deaths in the ICU include patients who were enrolled in the ICU or were admitted to the ICU after the diagnosis of a bloodstream infection.			
§ The length of stay in the ICU was evaluated in patients who were enrolled in the ICU or were admitted to the ICU after the diagnosis of a bloodstream infection.			
¶ Included are data for the patients who received vasopressors at any time after enrollment (722 patients in the 7-day group and 743 patients in the 14-day group).			
Included are the data for patients who received mechanical ventilation (469 patients in the 7-day group and 488 patients in the 14-day group).			
** Data regarding antibiotic-free days are missing for 2 patients in the 14-day group.			

Table 2. Detailed results for 90-day mortality, secondary clinical, resource, safety, and resistance outcomes.

Figure 2: Primary Outcome Analyses

All-Cause Mortality at 90 Days in ITT, PP, and mITT

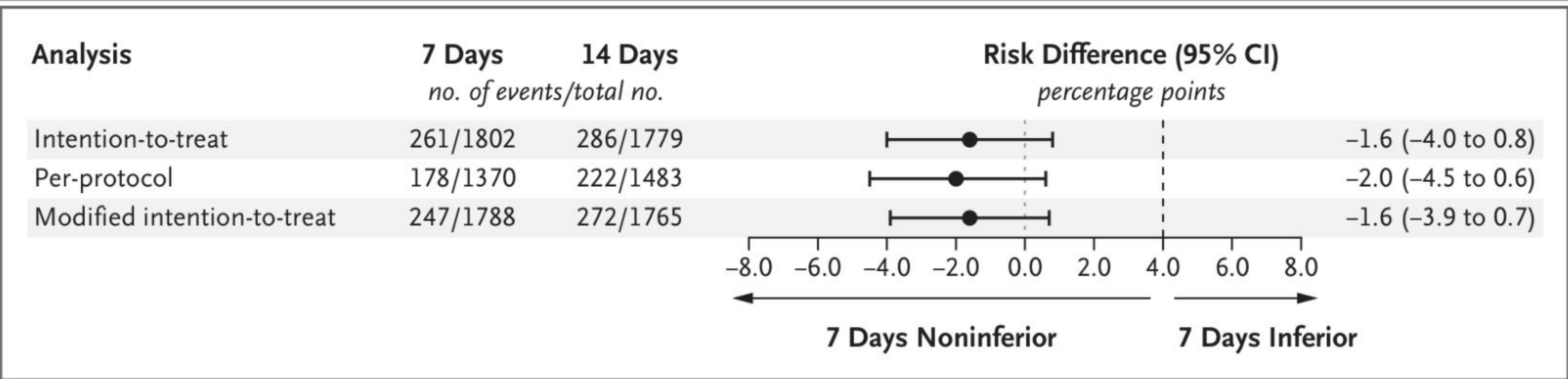


Figure 2. Primary Outcome According to Analysis.

Shown are the differences between the groups in the primary outcome — death from any cause by 90 days after the date of diagnosis of a bloodstream infection — in the intention-to-treat, per-protocol, and modified intention-to-treat analyses. The modified intention-to-treat analysis excluded patients who died before day 7 of treatment (i.e., before divergence in the treatment-duration assignment). A 95.7% confidence interval is shown for the intention-to-treat analysis (accounting for alpha spending in interim analyses), and 95% confidence intervals are shown for the other two analyses. The widths of the confidence intervals have not been adjusted for multiplicity. The dashed line indicates the noninferiority margin of 4 percentage points.

Figure 2. Intention-to-treat, per-protocol, and modified ITT analyses all exclude the 4.0% noninferiority margin.

Protocol Adherence and Deviations

8 days

median duration (7d)

14 days

median duration (14d)

23.9%

nonadherence in 7d

-2.0%

risk diff per-protocol

- Median Durations: 7-day group median was 8 days (IQR 7-11) vs. 14-day group median of 14 days (IQR 13-15).
- 7-Day Nonadherence: 23.9% total (23.1% received >9 days of antibiotics; only 0.8% received <5 days).
- 14-Day Nonadherence: 16.5% total (10.7% received >16 days; 5.8% received <12 days).
- Per-Protocol Result: Risk difference of -2.0% (95% CI: -4.5 to 0.6%) reinforces the ITT noninferiority result.
- Conclusion: Clinical deviations in duration did not undermine the robustness of the noninferiority claim.

Figure 3: Subgroup Analysis

Primary Outcome Across Prespecified Subgroups

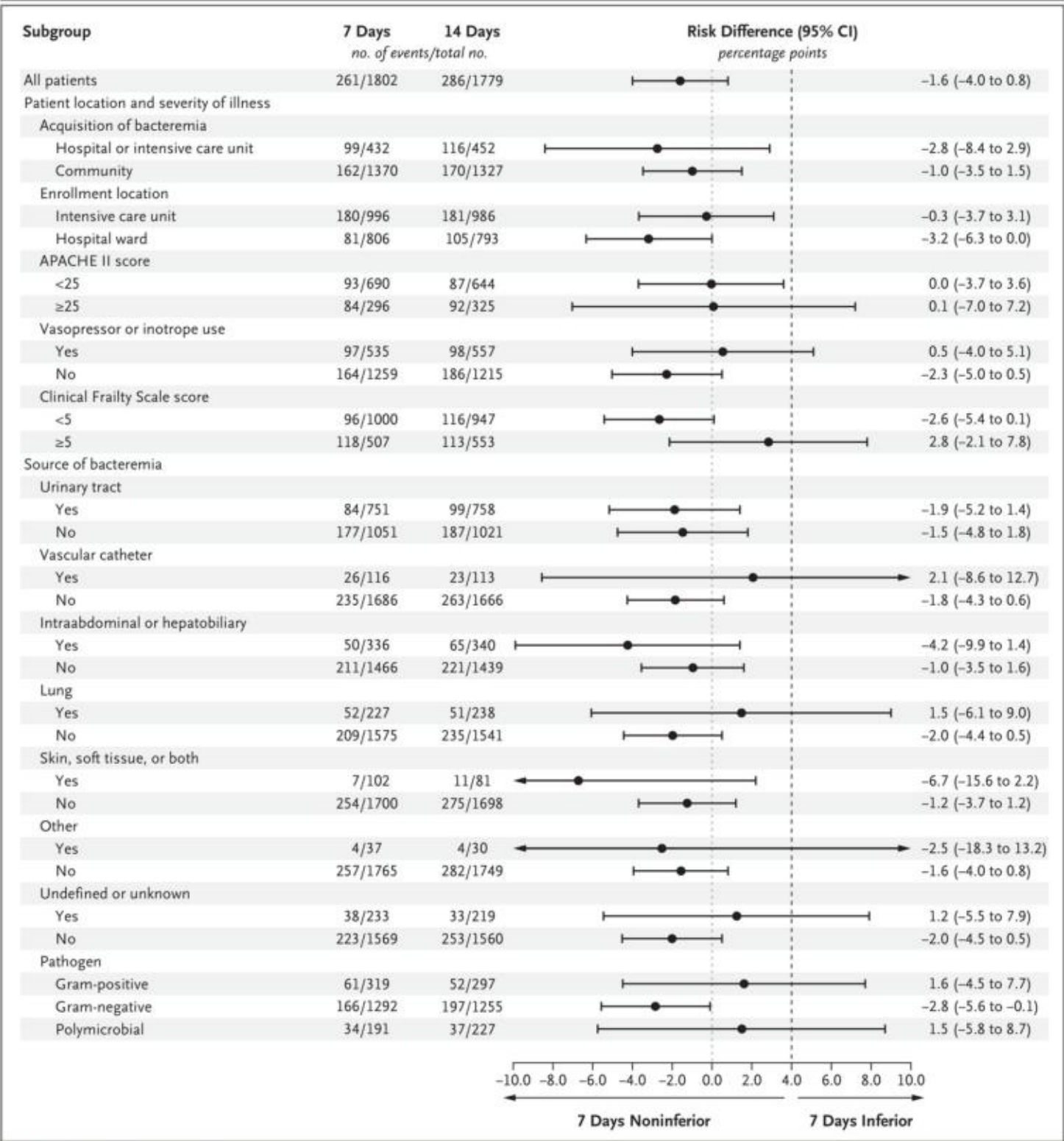


Figure 3. Primary Outcome According to Subgroup.
A 95.7% confidence interval (accounting for alpha spending in interim analyses) is shown for the overall population (the primary analysis), and 95% confidence intervals are shown for the subgroups. The widths of the confidence intervals have not been adjusted for multiplicity. The dashed line at 4 percentage points indicates the noninferiority margin. The Acute Physiology and Chronic Health Evaluation (APACHE) II score ranges from 0 to 71, with higher scores indicating an increased risk of death; the Clinical Frailty Scale score ranges from 1 to 9, with higher scores indicating greater frailty.

Figure 3. 90-day mortality risk differences across patient location, illness severity, source, and pathogens.

Subgroup Critique: Severe Illness & Pathogens

WHAT SUPPORTS CONSISTENCY

- ICU Cohort: High-severity patients in the ICU (55%) showed consistent noninferiority with 7 days of treatment.
- Shock Cohort: Patients receiving vasopressors at baseline showed no benefit from a 14-day duration.
- Source Consistent: Noninferiority held across urinary, catheter, abdominal, and lung bacteremias.
- Microbiology: Validated for both Gram-negative and Gram-positive bacteremia (excluding *S. aureus*).

PRECISION CAVEATS

- Clinical Caveat: Smaller subgroups have wide confidence intervals; clinical judgment remains necessary.

Stewardship Rule: Although subgroups support 7 days, clinical stability and source control must dictate final decisions.

Secondary Safety and Adverse Outcomes

1.7% vs 2.0%

C. difficile rates

9.5% vs 8.5%

secondary AMR infection

+5 days

antibiotic-free days

<1.1%

drug toxicity / AKI

- C. difficile Rates: Low and statistically similar (1.7% in 7-day vs. 2.0% in 14-day, Risk Difference -0.2%).
- AMR Colonization: Secondary resistant infections in hospital were similar (9.5% vs. 8.5%, Risk Difference 1.1%).
- Toxicity Measures: Drug allergy (0.8% vs 1.1%), anaphylaxis (0.1% vs 0.1%), and AKI (0.8% vs 0.9%) were low.
- Resource Benefits: Shorter duration yielded a median of 5 additional antibiotic-free days (19 vs. 14 days).
- Limitations: Safety monitoring relied on routine clinical ordering rather than active surveillance swabbing.

CRITICAL APPRAISAL

Internal Validity, Risk of Bias, GRADE, and Bayesian Reanalysis

Results Critical Appraisal

- Trial Strength: Blinding of outcome assessors protects against subjective measurement bias.
- High Retention: Over 99% complete follow-up at 90 days minimizes bias from missing data.
- Consistency: Alignment of ITT, per-protocol, and modified ITT results makes the conclusion highly defensible.
- Precision: The narrow 95.7% CI (-4.0% to 0.8%) strongly rules out a clinical mortality increase with 7 days.

Verdict: Excellent study integrity. Blinded outcome assessment, high follow-up completion (99.3%), and consistent ITT / per-protocol results provide robust evidence.

Risk of Bias 2 (RoB 2) Summary

Domain 1: Randomisation process Low Risk (Centralised Web-based randomisation with allocation concealment until Day 7)	LOW RISK
Domain 2: Deviations from intended interventions Low Risk (Allocation concealed; protocol deviations balanced and per-protocol consistent)	LOW RISK
Domain 3: Missing outcome data Low Risk (Only 0.7% lost to follow-up, complete 90-day vital status)	LOW RISK
Domain 4: Measurement of the outcome Low Risk (Blinded outcome assessors for secondary outcomes; objective primary mortality)	LOW RISK
Domain 5: Selection of the reported result Low Risk (Registered prospectively, pre-specified statistical analysis plan followed)	LOW RISK
OVERALL RISK OF BIAS JUDGEMENT	LOW RISK OF BIAS

GRADE Certainty of Evidence

GRADE FINAL CERTAINTY

HIGH CERTAINTY

We have high confidence that the true effect lies close to that of the estimate of the effect.

Outcome: All-cause mortality by 90 days

Risk of Bias: NONE

No downgrade. Multicenter, allocation-concealed, low risk of bias RCT.

Inconsistency: NONE

No downgrade. Results consistent across subgroups (wards/ICUs/pathogens).

Indirectness: NONE

No downgrade. Evaluates patient-important hard endpoint (90-day mortality) directly.

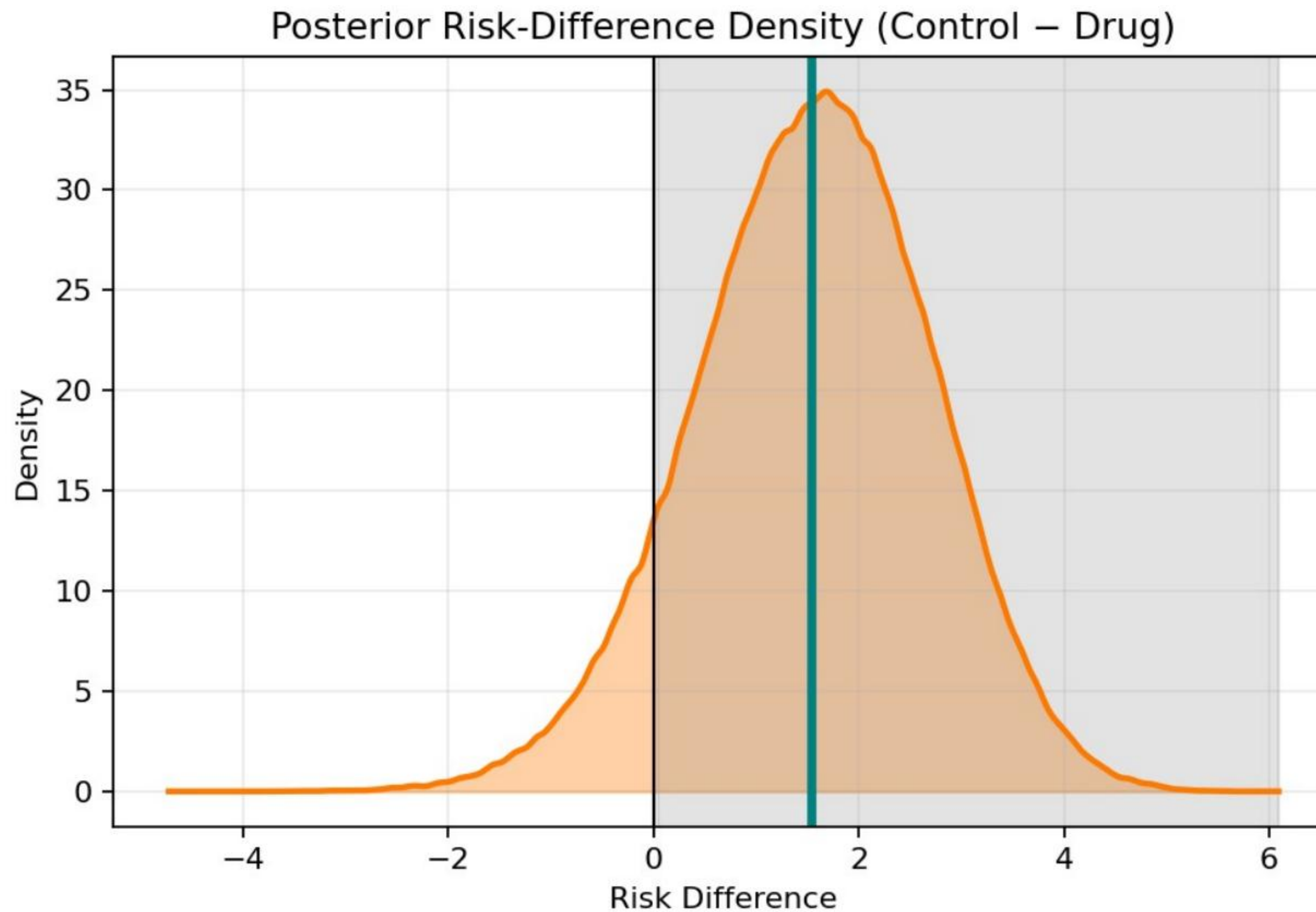
Imprecision: NONE

No downgrade. Large sample size (3,610 analyzed); CI excludes noninferiority margin.

Posterior Distribution: Risk Difference

Exploratory Bayesian Reanalysis of the Primary Outcome

BF10 (point-null $\theta=0$ vs Normal prior) = 0.221 | Mean RD=1.54pp | 95% CrI [-0.82, 3.69]pp



Posterior distribution of the risk difference (Control – Drug). Posterior mean is 1.53% in favor of 7 days.

Bayesian Reanalysis Interpretation

1.53%

posterior mean RD

90.6%

prob lower mortality

>99.9%

prob noninferiority

0.221

Bayes Factor null support

- Posterior Mean: Difference is 1.53% in favor of the 7-day course (95% Credible Interval: -0.83% to 3.70%).
- $P(RD > 0)$: 90.6% probability that a 7-day course leads to lower mortality than a 14-day course.
- $P(RD > -4\%)$: >99.9% probability that 7 days is not worse than 14 days by more than the 4% margin.
- Bayes Factor (BF10): 0.221 (strong evidence supporting the null hypothesis of no difference).
- Takeaway: Bayesian sensitivity analysis strongly supports the trial's frequentist noninferiority claim.

Figure S1: Meta-Analysis of Trial Evidence

Combining BALANCE with Prior Randomized Controlled Trials

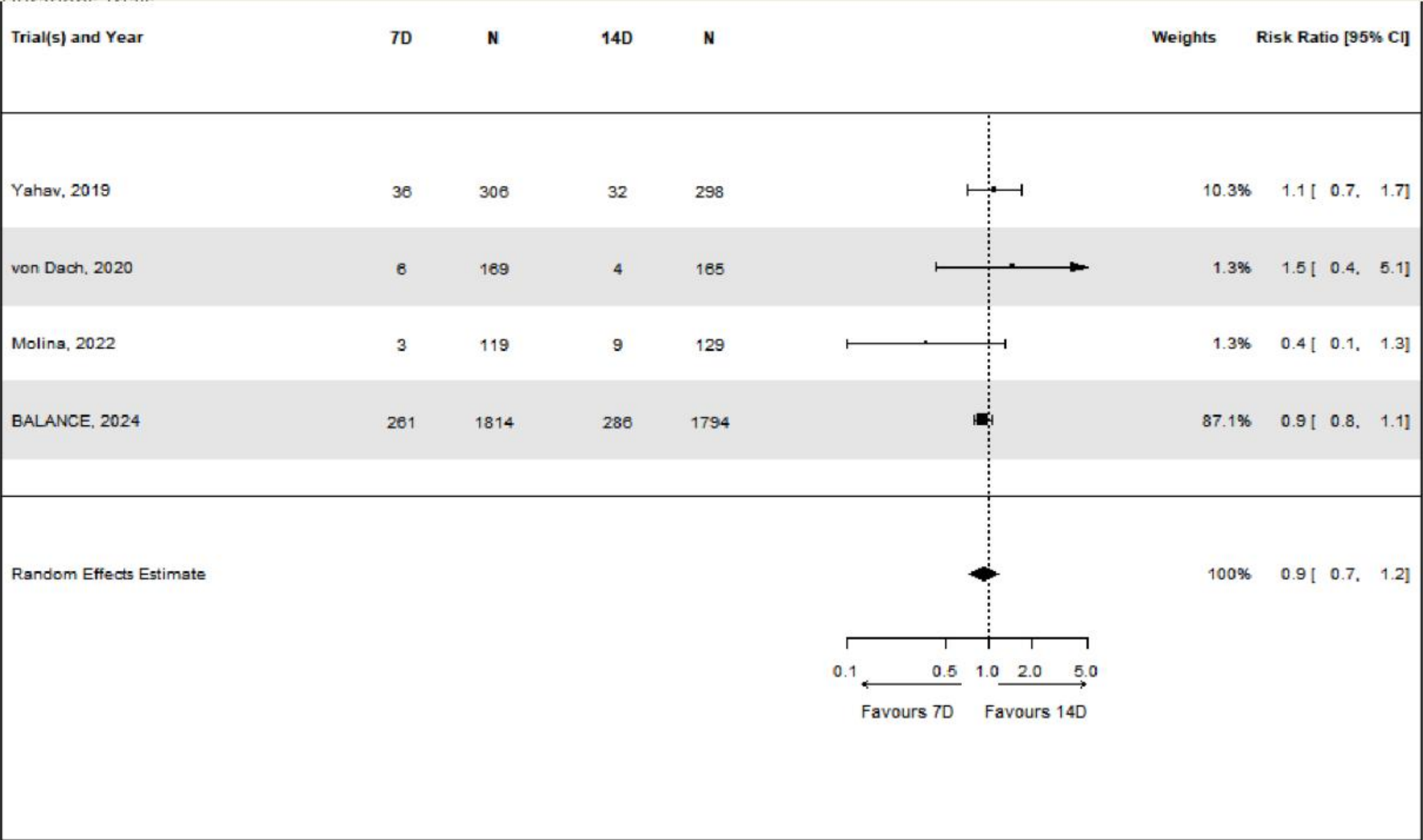


Figure S1: Meta-analysis combining BALANCE RCT results with other published randomized trials comparing mortality in patients receiving 7 versus 14 days of antibiotic treatment for non-neonatal bloodstream infections (Random effects, Inverse variance weighting). The BALANCE RCT contributes 87.1% of the meta-analysis weight. $I^2=0\%$. Test for Heterogeneity $Q(df = 3) = 3.2113, p= 0.36$.

Figure S1. Meta-analysis combining BALANCE with Yahav et al. (2019), von Dach et al. (2020), and Molina et al. (2022).

APPLICABILITY & TAKEAWAYS

Local clinical use, hospital guidelines, and discussion

Local Applicability: North Indian Patients

STEWARDSHIP & FINANCIAL BENEFITS

- High MDR Burden: Local prevalence of ESBL and CRE in Ghaziabad makes antibiotic stewardship critical.
- Stewardship Tool: Halving duration reduces selection pressure, combatting further resistance.
- Financial Benefit: Halving antibiotic course directly reduces out-of-pocket costs, minimizing financial toxicity.

CRITICAL SAFETY PREREQUISITES

- Initial Adequacy: 7 days is safe ONLY if the initial empiric antibiotic is active against the pathogen.
- Source Control Priority: Source control must be confirmed; if absent, short-course treatment will fail.

Ghaziabad Clinical Context: Stewardship is crucial due to high baseline ESBL/CRE rates, but initial empiric coverage must be active.

Clinical Takeaways for Wards & ICUs

ADOPT 7-DAY DEFAULT

- Standard of Care: Adopt a default 7-day duration for stable, source-controlled bacteremia.
- Cost & Resource Benefits: Shorter courses reduce drug costs, ICU stay, and HAI exposure.

ABSOLUTE CLINICAL EXCLUSIONS

- Critical Pathogens Excluded: Do NOT apply 7 days to *S. aureus*, *S. lugdunensis*, or fungemia.
- Immunosuppressed Exclusions: Severely neutropenic or post-transplant patients require longer courses.
- Source Control Rule: Ensure catheters are removed or abscesses drained before stopping at day 7.

Practice Bottom Line: Short-course therapy (7 days) is safe and effective for stable, source-controlled bacteremia.

Journal Club Discussion Questions

SOURCE CONTROL

1. In our hospital, how quickly can we confirm urological or radiological source control for bacteremic patients?

PERSISTENT SYMPTOMS

2. How do we manage patients who are stable but still have mild leucocytosis or low-grade fever on Day 7?

ICU CHECKLISTS

3. How can we implement a stewardship checklist in our ICU to enforce a 7-day cutoff for eligible patients?

LOCAL VERIFICATION

4. Should we collect local data on relapse rates if we implement this 7-day standard at Santosh Hospital?